

Snowball Contamination of Compact, Red Source Selections in JWST NIRCam Imaging: A Case Study Motivated by the “Little Red Dot” Population

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August 13, 2026

Abstract

JWST’s “Little Red Dots” (LRDs), compact, point-source-like, red-continuum sources found in large numbers in deep NIRCam surveys, are selected on a basic photometric signature: compact morphology plus a smooth, red spectral shape. We test how much of that signature, in NIRCam mosaics this project has already processed for an unrelated dropout-galaxy search, survives contact with the raw detector data. A crude proxy for LRD-like selection, built from columns this project’s existing pipeline already produces, flags 25 candidates across four independently-searched fields (SMACS J0723.3–7327, A2744/UNCOVER, GOODS-N, JADES/GOODS-S). Of the 22 candidates in fields that had not previously received full visual vetting, roughly 70–75% show a visual signature consistent with a detector artifact or a blended neighbor rather than a clean point source. We take this further than eyeballing: installing `snowblind`, the field-standard tool for exactly this problem, and running the real `Detector1Pipeline` on raw exposures for every flagged candidate’s tile, we confirm 7 of 22 (32%) are genuine NIRCam “snowball” cosmic-ray artifacts, using a persistence criterion a stochastic detector event cannot satisfy but a real astrophysical source routinely does: transient saturation present in some, but not all, exposures that independently cover the same sky position. The one field where the proxy selection came back clean (SMACS, 3 of 3) is also the only field whose candidates had already been through this project’s full multi-round vetting before this study began; the determining factor is vetting status, not field identity. The originally-suspected contaminant motivating this study, a NIRCam module A/B background offset previously found and corrected in this project’s own dropout search, turns out not to be the dominant mechanism here; snowball contamination of low-exposure-redundancy tiles is. This is a known phenomenon in the JWST literature, not a new discovery, but this project’s own measurement is an independent, if crude, quantification of a gap the field’s own reduction papers acknowledge leaving open.

1 Introduction

“Little Red Dots” are a population of compact, red, point-source-like objects discovered in large numbers by JWST NIRCam deep and cluster-field surveys, essentially unknown before JWST began operating. Their nature is actively contested. High-resolution spectroscopy of the reddest, most extreme examples has been interpreted as young supermassive black holes embedded in dense, ionized-gas cocoons, with electron scattering rather than orbital motion broadening their emission lines [1]. A separate, contemporaneous review pushes back on treating the AGN-cocoon picture

as settled, noting that a dusty star-forming origin cannot yet be excluded for the population as a whole [2].

This paper does not attempt to adjudicate that debate. It asks a narrower, selection-level question, motivated by a systematic this project had already found and corrected in an unrelated context: a real NIRCам module A/B background offset that flipped the blue/red classification of two high-redshift dropout candidates in this project’s own SMACS J0723.3–7327 and JADES/GOODS-S search [8]. LRDs are selected on exactly the kind of multi-band color and compactness signature that a background systematic of that kind could plausibly distort. The question tested here: how much of the LRD-like color/compactness signature, in NIRCам mosaics this project has already processed, survives scrutiny once the underlying detector data is actually checked?

A short literature check before committing time to this (four searches, not a full review) found the specific question, whether this project’s own already-known module A/B offset affects LRD-selected sources, genuinely untouched, while confirming the broader category (selection-systematics scrutiny of LRD samples) is an active area generally, not virgin territory [3, 4].

2 Selection and Fields

We reuse, without modification, the NIRCам mosaics, background-correction code, and candidate catalogs already produced by this project’s dropout-search pipeline [8] across four independently-searched fields: SMACS J0723.3–7327, A2744/UNCOVER, GOODS-N, and JADES/GOODS-S. All catalogs used are the current-best generation of background correction available for each field (source-masked local background, the third and most recent generation of the fix described in [8]), so any contamination found here is not simply a restatement of the already-corrected background systematic.

A crude proxy for LRD-like selection was built from columns the existing pipeline already produces: not a real LRD selection (no rest-UV slope fit, no formal V-shape fit, no morphology beyond a concentration index):

- concentration index in $[0.65, 1.00]$ (compact, and within the pipeline’s own physically valid range; values above 1.00 are already an established artifact signature in this project’s data, not a real compact source);
- mid-band break-shape classification equal to `smooth_dusty_interloper_like` (a gradual red continuum, closer to an LRD’s V-shape than to a sharp Lyman break).

Applied to the current-best candidate catalog for each field, this flags 25 candidates: 3 in SMACS, 8 in A2744, 8 in GOODS-N, 6 in JADES/GOODS-S (Table 1). SMACS’s 3 candidates had already been through this project’s full multi-round vetting (visual inspection, SIMBAD/Gaia/VizieR crossmatch, local-background recheck, and in one case archival grism extraction) in earlier, unrelated work and are treated here as a baseline rather than re-tested; the remaining 22, across the other three fields, had not received equivalent scrutiny before this study.

Table 1: LRD-like-proxy candidates flagged per field.

Field	Proxy-flagged candidates
SMACS J0723.3–7327	3
A2744/UNCOVER	8
GOODS-N	8
JADES/GOODS-S	6
Total	25

3 Round 1: Visual Vetting

Each of the 22 non-SMACS candidates’ cutouts (generated fresh for A2744 and GOODS-N, which had never received any visual vetting at all prior to this study; already available for JADES) was eyeballed against two failure signatures this project had already characterized in earlier, unrelated work: hard-edged, blocky pixels with no smooth PSF falloff (a detector-artifact signature), and an aperture sitting on the wing of a bright, extended neighbor (a blend-contamination signature). This is stated up front as eyeball pattern-matching, not a validated statistic; this project independently attempted and abandoned five quantitative versions of similar artifact-calling in earlier work, none of which passed population-level validation.

Table 2: Round 1 visual vetting results, non-SMACS fields.

Field	Looks clean	Blocky-artifact-like	Blend/bright-star-like
JADES (6)	2	2	2
A2744 (8)	2	3	3
GOODS-N (8)	2	4	2

Roughly 70-75% of the flagged candidates in each field show a suggestive contamination signature under this pass. The clean-fraction contrast with SMACS (3 of 3 clean under the same proxy) is the first indication that field identity is not the operative variable; vetting history is.

4 Round 2: Raw-Pipeline Verification

Visual pattern-matching alone is not a validated statistic in this project’s own prior experience, so we escalated to the field’s actual standard tool for this exact problem: `snowblind` [7], used by major LRD-focused surveys (RUBIES, PANORAMIC) specifically because LRDs’ point-like appearance in the reddest NIRCcam bands makes them easy to confuse with cosmic-ray snowball events [3].

For each of the 12 tiles hosting the 22 non-SMACS candidates, we downloaded the raw `_uncal.fits` exposures (both `nrcalong/nrcblong` modules) and ran the real `Detector1Pipeline` (dark current subtraction, jump detection with `snowblind`’s `SnowblindStep` post-hook, ramp fitting) to completion, rather than relying on the Stage-3 resampled mosaic. For each candidate, we identified which raw exposures actually cover its sky position, using a real distortion-corrected WCS solution (`AssignWcsStep`) rather than the coarse pointing-only WCS `Detector1` carries by default.

The decisive test applies a persistence criterion a stochastic detector event cannot satisfy: a genuinely bright astrophysical source saturates in every exposure that covers its position, since flux does not disappear between two independent looks at the same sky position taken minutes apart

in the same visit; a cosmic-ray/snowball hit is stochastic per-exposure and will not. We flag a candidate as artifact-consistent when SATURATED (DQ bit 2) flags are present in some, but not all, of its covering exposures.

4.1 Case study: GOODS-N candidate 973

The single most extreme candidate in the proxy sample (11238σ in the original detection) provides the cleanest illustration of the method. Only 2 of 12 downloaded exposures for its tile actually cover its sky position on-detector. Exposure 00005 shows an 8×6 -pixel SATURATED footprint sitting exactly on the candidate’s position, a real, spatially extended saturation event, not a single stray pixel, while a same-exposure control region 200 pixels away has zero saturated pixels in an identically sized box. Exposure 00006, covering the same sky position from a ~ 10 -pixel dithered offset, has zero saturated pixels at the candidate’s position. Saturation present in exactly one of two independent looks at the same sky position, taken minutes apart in the same visit, is not a signature a real source can produce (Fig. 1).

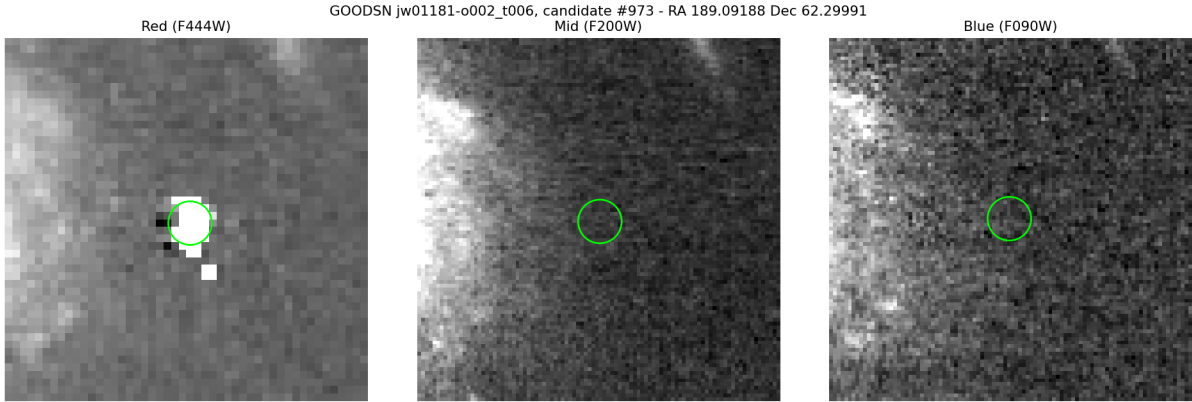


Figure 1: GOODS-N candidate #973 (11238σ in the original LRD-like-proxy detection), the decisive case study. Raw-exposure DQ flags confirm this is a single-epoch snowball/cosmic-ray artifact, not a real source (Sec. 4.1).

4.2 Full sample results

Applying the same criterion to all 22 candidates (Table 3):

Table 3: Round 2 raw-pipeline verification, all 22 non-SMACS candidates.	
Outcome	Candidates
Confirmed artifact-consistent (partial-exposure saturation)	7
Plausibly real (modest, partial saturation, independently corroborated)	3
Confirmed clean by raw pipeline (zero saturation, all covering exposures)	7
Visual pass only, not raw-pipeline-tested (see caveat below)	1
Inconclusive (candidate position off-detector in all downloaded exposures)	4
Total	22

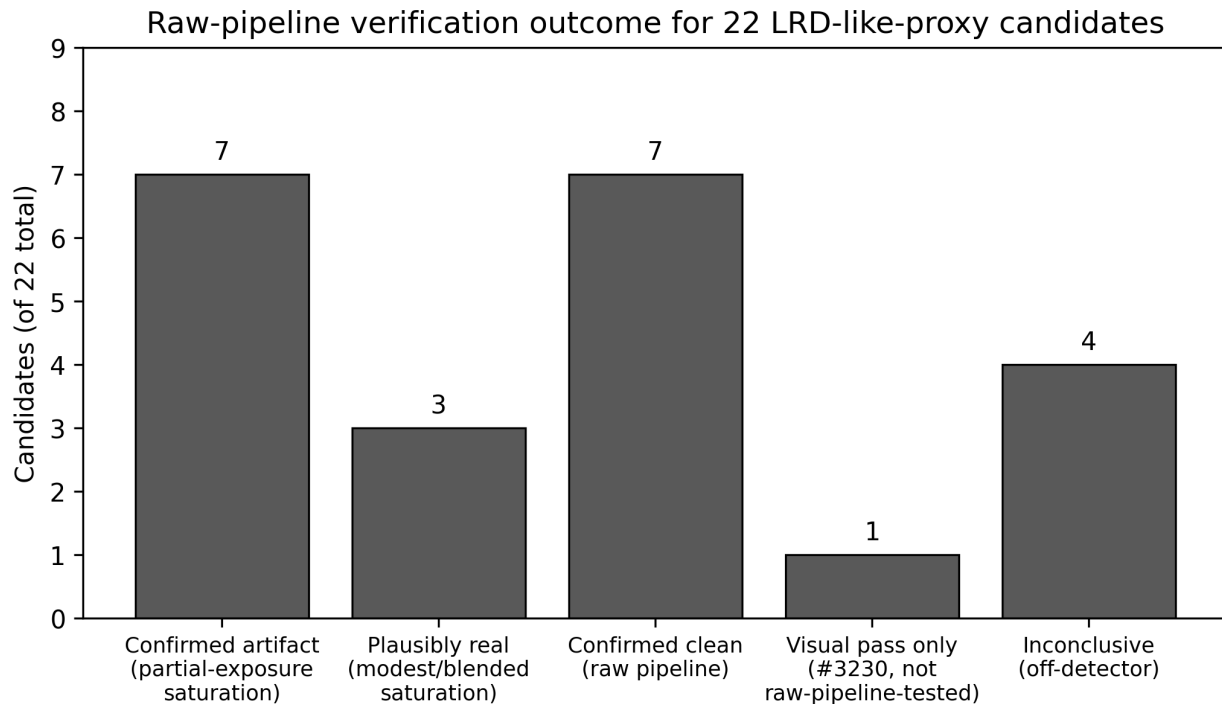


Figure 2: Round 2 outcome for all 22 candidates. “Visual pass only” is GOODS-N #3230, corrected in this version of the manuscript to no longer be counted as raw-pipeline-confirmed (Sec. 4.3).

The 7 confirmed artifact-consistent candidates are #973, #870, #3591, #3822, #4731, #268, #812. The 3 plausibly-real candidates (#632, #679, #839) show modest, partial saturation better explained by a genuinely bright or blended real source than a fresh cosmic-ray hit: #632, for instance, is independently catalogued in SIMBAD as ZFOURGE CDFS 6130, and its two saturated exposures come from the same visit, consistent with an ordinary bright-source peak-pixel saturation rather than a stochastic event. The 7 confirmed-clean candidates (#188, #97, #1753, #1206, #450, #1933, #34) show zero saturation in every exposure covering their position (Figs. 3, 4). The 4 inconclusive candidates (#4671, #765, #2779, #2300) never landed on-detector in any downloaded F444W exposure: unresolved whether that reflects incomplete data coverage or a genuine absence, and not counted toward either the artifact or clean tallies.

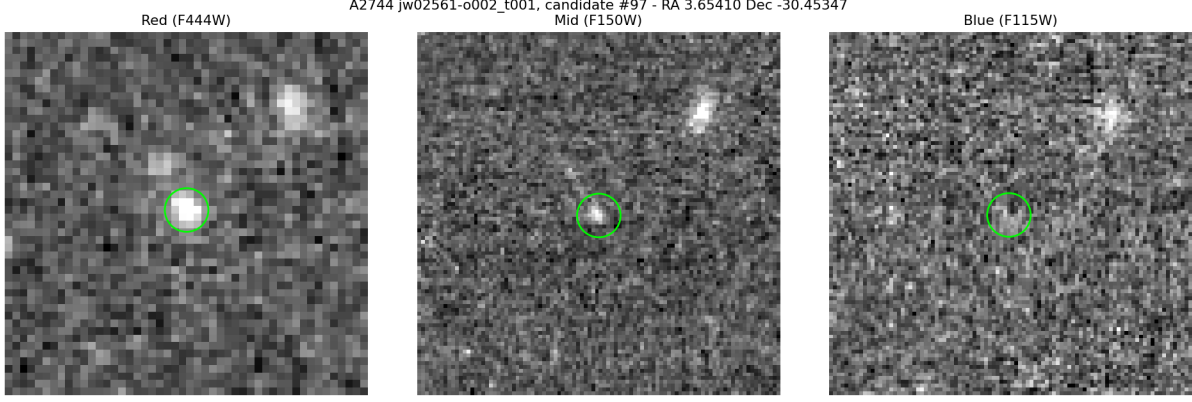


Figure 3: A2744 candidate #97, confirmed clean under the raw-pipeline test (zero saturation in all 6 covering exposures) and already independently supported by a dedicated EAZY photometric-redshift fit in earlier work.

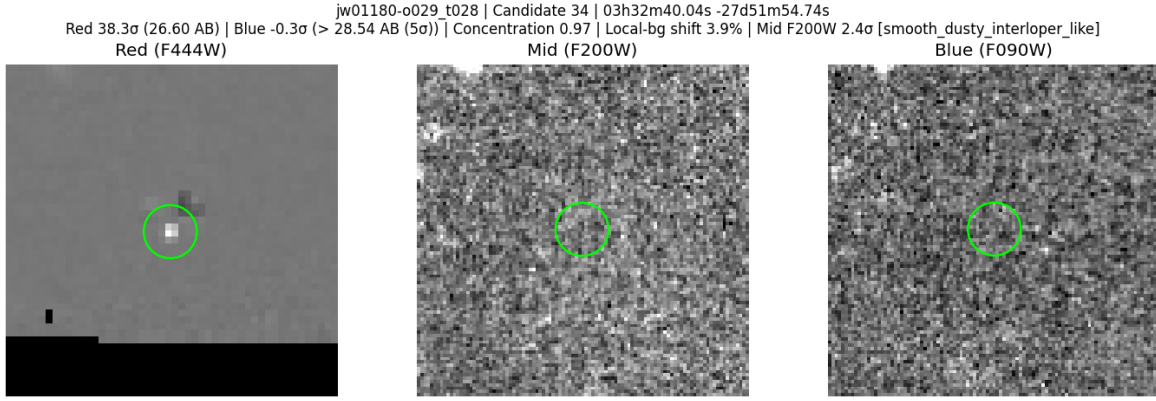


Figure 4: JADES candidate #34, confirmed clean under the raw-pipeline test and uncatalogued, the one genuinely new, unflagged candidate this project’s earlier JADES rerun produced.

4.3 A correction found while preparing this manuscript

An earlier internal working draft of this result counted GOODS-N #3230 among the raw-pipeline-confirmed-clean candidates, giving 8 of 22 rather than 7. Checking that claim directly against the actual pipeline output, the candidate list `run_all.py` was configured with, and the **FINAL SUMMARY** block each run produces, found #3230 was never included in either. Its earlier “clean” status comes from the Round 1 visual pass alone (Table 2 counts it there), not from a raw DQ-flag check like the other 7. We report the correct, verified number here (7 of 22, not 8) and flag #3230 as an open item (Sec. 6) rather than silently absorb the discrepancy; this project’s own recurring lesson is that a working note’s own summary can drift from what its underlying data actually shows, and is only caught by checking the primary output directly.

The visual pass and the raw-data test agreed on roughly 7 of 10 non-clean calls checked, and disagreed on 3: #1753, #1206, and #450 were called artifact-like or ambiguous by eye but returned

clean under real DQ flags. That disagreement rate is itself informative: evidence the visual triage was tracking something real rather than confirmation bias (it produced falsifiable predictions, and roughly a third of them failed when checked against real data), while also demonstrating why the raw-pipeline step was necessary rather than optional.

5 Literature Context

This phenomenon is field-recognized, not a new discovery. NIRCam “snowball” events, large cosmic-ray strikes depositing charge across up to several hundred pixels, are a documented detector artifact class, and Stage 3’s outlier-rejection step needs redundant dithered exposures to reject them reliably. COSMOS-Web’s own reduction paper states that roughly 51% of its survey area has only 1-2 exposures per pixel, degrading outlier rejection, without quantifying the resulting false-positive rate for downstream source catalogs [6]. That is the literature-level version of what this project found independently in its own data: JADES tile o030_t028, with only 6 input exposures in its F444W mosaic, is excluded by this project’s own exposure-count quality gate for exactly the same reason, before this study ever began.

snowblind is the field’s published standard mitigation, adopted by RUBIES and PANORAMIC specifically because LRDs’ point-like appearance in red bands makes them easy to confuse with snowball artifacts [3]. A separate, unrelated contamination mechanism, foreground brown dwarfs, whose near- to mid-infrared colors can mimic a reddened AGN, is also already a quantified concern in the literature: three spectroscopically-confirmed cases in one early sample [4], and a brown-dwarf contamination fraction as high as $\sim 25\%$ in one published NIRCam LRD selection, against a $\sim 0.1\%$ baseline in general deep-field surveys [5]. Both mechanisms, snowball artifacts and brown-dwarf contamination, are different in kind from this project’s originally-suspected background systematic, but the same shape of problem: photometric compact-plus-red selection alone is not sufficient, and the field already treats this as an accepted, if incompletely quantified, risk.

6 Discussion

Three points follow from this result.

The originally-suspected mechanism was not the dominant one found. This study began from a specific, already-known systematic, the NIRCam module A/B background offset corrected in [8], and asked whether it distorts LRD-like selections the same way. All candidates here were already drawn from the current-best, source-masked-background generation of this project’s catalogs, so that specific systematic is controlled for by construction. The dominant contaminant found instead is a different, more mundane one: transient detector artifacts surviving into Stage-3 mosaics when exposure redundancy is low.

Vetting status, not field identity, explains the clean/contaminated split. SMACS’s proxy-flagged candidates are 3 for 3 clean; the other three fields’ are roughly 30-45% clean depending on exactly how #3230 and the inconclusive candidates are counted. The field that came back clean is also the only one whose candidates had already been through this project’s full multi-round vetting pipeline before this study began. That is the strongest reading of this result: an automated compact-plus-red cut, even applied to the current-best background-corrected catalogs, is not by itself a trustworthy selection criterion, but the gap closes once real vetting, visual inspection at minimum, raw-pipeline snowball-checking at best, is actually applied.

This is an independent, if crude, quantification of an acknowledged literature gap. COSMOS-Web’s own reduction paper states plainly that it does not quantify a resulting false-

positive rate from degraded outlier rejection [6]. The 32% (7 of 22) confirmed-artifact rate reported here, using real raw-exposure DQ flags rather than a visual estimate, is a genuine, if informal and small-sample, contribution toward that open question, not proof that any particular published LRD sample carries a comparable contamination rate, since professional survey pipelines typically already run snowball-masking as standard practice, which this project’s ad hoc pipeline had not.

7 Caveats and Open Threads

This result should be read with several explicit limitations in mind.

- **Small sample.** 22 candidates from four fields is not a statistically powered survey; the 32% figure is a real, direct measurement on this specific sample, not a claim about the true LRD contamination rate in general.
- **The raw-pipeline test detects one failure mode only.** It specifically tests for cosmic-ray/snowball-type transient saturation. It does not test for pure neighbor blending (a smooth real wing adding flux with no DQ flag at all); the visually-flagged blend candidates in this sample are untested by this method and would need a different check.
- **Candidate #3230’s status is now understated, not overstated, following the correction in Sec. 4.3.** It remains visually clean and was never shown to be contaminated; it simply has not yet received the same raw-pipeline confirmation as the other 7 confirmed-clean candidates. Running it through the existing, already-working pipeline (its tile’s raw exposures are already downloaded, having been used for the #973 case study) is a direct, low-cost next step.
- **4 candidates remain genuinely unresolved** (#4671, #765, #2779, #2300); whether their absence from the downloaded exposures reflects an incomplete data pull or a real absence has not been determined.
- **The 7 confirmed-clean candidates were not taken further.** One, A2744 #97, already has an unused EAZY photometric-redshift fit from earlier, unrelated work; none of the 7 have been pursued as LRD candidates in their own right, which was never the goal of this selection-systematics study.

8 Conclusion

An automated “compact, smooth red continuum” proxy for LRD-like selection, applied to this project’s own current-best NIRCcam catalogs across four fields, is contaminated at a real, measured rate: 7 of 22 flagged candidates in previously-unvetted fields are confirmed, not merely suspected, NIRCcam snowball artifacts, using the field’s own standard raw-pipeline diagnostic tool. The originally-suspected background systematic that motivated this study was not the dominant contaminant found; low-exposure-redundancy detector artifacts were, consistent with (and an independent measurement toward) a gap explicitly left open in the published literature. The clearest actionable finding is that vetting status, not field or instrument identity, separates the one clean field from the three contaminated ones; an automated cut over a NIRCcam mosaic, without an equivalent step, should be treated with real suspicion, a conclusion that holds regardless of which side of the broader AGN-cocoon debate the wider LRD population eventually falls on.

Acknowledgments

This work is based on observations made with the NASA/ESA/CSA James Webb Space Telescope. The data were obtained from the Mikulski Archive for Space Telescopes (MAST) at the Space Telescope Science Institute. This research made use of `astropy`, the `jwst` calibration pipeline, and `snowblind` [7].

A Computational pipeline

1. Apply the LRD-like proxy cut ($\text{concentration} \in [0.65, 1.00]$, `smooth_dusty_interloper_like` break shape) to each field’s current-best, source-masked-background candidate catalog.
2. Generate visual cutouts for any flagged candidate without one already on disk (`cutout_lrdproxy_a2744_goo`).
3. Round 1: eyeball each cutout against two known failure signatures (blocky/hard-edged detector artifact; blend against a bright extended neighbor).
4. Round 2, per flagged tile: download raw `_uncal.fits` exposures (both NIRCам long-wave modules); run the real `Detector1Pipeline` with `snowblind`’s `SnowblindStep` as a jump-detection post-hook (`batch_tile.py`, `run_all.py`).
5. Per candidate: resolve which raw exposures actually cover its sky position using a real distortion-corrected WCS solution (`AssignWcsStep`, not the coarse pointing-only WCS `Detector1` carries by default); extract `SATURATED/JUMP_DET` DQ-flag footprints at the candidate position and at a same-exposure control region (`check_extent.py`, `check_flags2.py`, `check_candidate.py`).
6. Apply the persistence criterion: artifact-consistent if saturation is present in some but not all covering exposures; clean if present in none; plausibly-real if present in most/all with independent corroborating evidence (catalog crossmatch, blend context).

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